
When Shift Happens - Confounding Is to Blame

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Abstract

Distribution shifts introduce uncertainty that undermines the robustness and generalization of machine learning models. Conventional wisdom holds that learning causal-invariant representations improves robustness, yet recent empirical studies report a counterintuitive pattern: (i) empirical risk minimization (ERM) rivals or outperforms state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) its OOD performance improves when *all* available covariates—not just causal ones—are used. **This paper explains that pattern and, in doing so, makes four contributions.** **First**, we identify *hidden confounding* as the culprit: shifts in hidden confounders induce data-distribution changes that violate the assumptions underlying existing OOD approaches. **Second**, under such confounding shifts we prove that effective generalization *requires* learning environment-specific relationships—invariant representations, while necessary, are provably insufficient. **Third**, we show that augmenting models with proxies for hidden confounders provably increases the predictive information available and thus mitigates confounding shift, which motivates principled covariate selection. **Fourth**, experiments on eight real-world tabular benchmarks and synthetic data confirm all three claims. Together, these results explain the surprising strength of ERM under real-world shifts and offer concrete theoretical and practical guidance for designing robust OOD generalization algorithms.

1 Introduction

Generalization—the ability to draw reliable conclusions about unseen data based on observed data—is central to numerous scientific fields. In medicine and the social sciences, it is embodied as *external validity*, ensuring findings from one population are applicable to a different population [13]; in ecology, it supports *space-for-time substitutions*, where spatial variation is used as a proxy for temporal change to infer long-term ecological patterns [60]; and in engineering, it drives *robust control*, where models must maintain performance in the presence of unmodeled disturbances [35]. In recent years, machine learning has become a powerful tool for learning generalizable models [100, 88, 49].

Generalizing well requires robustness to distribution shifts between training and test data. A model trained to predict *food stamp reciprocity* from *household attributes* in one state, for instance, should still perform well when deployed in another. Such a model is said to achieve *out-of-distribution (OOD) generalization* when it maintains performance on both in-distribution (ID) data—from which training data are sampled—and out-of-distribution (OOD) test data. Over the last decade, whole families of methods have been proposed to secure provable OOD generalization under specific assumptions: domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94]. However, two findings sit uneasily with this view. **First**, under careful model selection [30, 38, 87], models based on standard *empirical risk minimization (ERM)* [86] often match the OOD generalization performance of purpose-built methods across a range of real-world applications [30, 24, 50, 57, 67, 87]. **Second**, a recent empirical study [57] on 16 real-world tabular datasets concludes that incorporating *all* available covariates when predicting the outcome—regardless of whether they directly affect it—can *improve* OOD generalization. Both findings contradict prevailing assumptions in the field. Explaining why they hold is the central question of this paper.

To build toward our answer, we first recall why invariance became the dominant lens. Distribution shifts are commonly observed when data originate from different environments [71]. For example, *food stamp reciprocity* may differ across states, because each state operates under different eligibility rules, yielding distinct, environment-specific distributions. In such data, certain statistical relationships between covariates and the outcome stay consistent across all environments—so-called *invariant relationships* [6]—and identifying them is widely regarded as the route to OOD generalization [6, 59, 31, 66, 56, 98, 63]. A special case is *causal invariance*, where the causal relationships in the data stay invariant across environments [71, 59, 6]. The key gap we exploit is this: in many real-world settings, not all covariates relevant for predicting an outcome are observed, due to limitations in data collection, privacy constraints, or measurement errors [14, 52, 20, 48]. Their absence disrupts precisely the invariant relationships that generalization relies on. The problem is most acute when the unobserved variables are *confounders* that influence both the observed covariates and the outcome. Such hidden confounders are pervasive, and shifts in their distributions define distinct environments. Ignoring these shifts not only degrades generalization [40, 4, 85, 62], but can drive models to learn incorrect relationships between observed covariates and the outcome.

Despite its importance, the impact of hidden confounding shifts on OOD generalization remains underexplored [4, 40, 85, 62]. This paper fills that gap, offering theory and empirical evidence that jointly explain the recent findings above [30, 24, 50, 57, 67, 87]. **We make the following four contributions:**

- **(C1) Confounding shift as the mechanism.** We motivate, from a causal perspective, why hidden confounding shift must be studied in OOD generalization (§ 3), and use it to explain why adding informative, non-causal covariates can help.
- **(C2) Invariance is necessary but insufficient.** We prove that maximizing the predictive information between model predictions and true outcomes *demands* explicitly learning environment-specific relationships; invariant representations alone, though necessary, are provably insufficient (§ 4.1, § 4.2).
- **(C3) Proxies for confounders help—provably.** We show that variables informative of the outcome or of hidden confounders increase the predictive information between predictions and true outcomes, establishing why principled covariate selection matters under hidden confounding shift (§ 4.3).
- **(C4) Empirical confirmation.** On real-world and synthetic data we show that (i) hidden confounding is prevalent in real-world tabular benchmarks, (ii) learning environment-specific relationships correlates positively with ID and OOD test accuracy, and (iii) incorporating informative, non-causal covariates improves generalization (§ 5).

In short, our work explains the surprising effectiveness of ERM under real-world distribution shifts and, more broadly, exposes the subtle yet inevitable role of hidden confounding in OOD generalization—making the case for methods that confront it head-on.

2 Related work

Out-of-distribution (OOD) generalization spans several subfields: domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. They differ in their assumptions: domain adaptation assumes access to unlabeled test data, domain generalization relies on environment labels during training, federated learning targets communication and privacy constraints across environments, and OOD detection flags samples that differ sharply from the training distribution. What unites them is a shared goal—learning invariant relationships. Yet a growing body of work argues that invariance alone is not enough and that additional inductive biases are needed [46, 72, 93]; in particular, Ye et al. [93] show that feature invariance is necessary but insufficient and stress the role of feature *informativeness*. Our contribution C2 sharpens this line of work: we study invariance and informativeness *together* and pinpoint hidden confounding shift as the setting where invariance provably fails.

Proxy variable adjustment. When confounding variables are observed during training but unobserved at test time, Landeiro and Culotta [40] adjust for confounding shifts to improve classifier performance. Building on proxy-based adjustment for causal effect estimation [55, 39], recent domain adaptation methods recast unknown distribution shifts as a causal effect identification problem [4, 85], and OOD generalization under hidden confounding shift has been studied under an overlapping-confounder-support assumption [62]. These works propose *how* to adjust; our contribution C3 is

complementary and explanatory—we characterize *why* proxy variables enhance OOD generalization, quantifying their effect through the predictive-information decomposition.

All-variable models vs. causal models. Nastl and Hardt [57] introduced a benchmark that partitions covariates into four groups—causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates—and showed that, across 16 datasets, models using all covariates Pareto-dominate those restricted to causal or arguably causal subsets on both ID and OOD data. Theoretical explanations for this behavior are scarce, and supplying one is exactly what our contributions C1 and C3 deliver. Related efforts partly address the question: for linear causal models, anchor regression [68] interpolates between all-covariate and causal-only estimation, while Eastwood et al. [21] show that unstable covariates can boost performance when they carry label information and are conditionally independent of the stable covariates given the label, adjusting the shift by inspecting the unlabeled test domain. Even so, on a medical dataset not built for this problem [8], ERM remains competitive with their method, echoing Nastl and Hardt [57]. Such results reflect a broader fact: under *well-specified* covariate shifts, maximum likelihood estimation attains minimax optimality for OOD generalization [26]. The distinguishing premise of our work is that real-world settings are rarely well-specified, precisely because of the hidden confounding shift we analyze.

3 Manifestations of hidden confounding shift

This section develops contribution C1. We first give the background needed to define hidden confounding shift, and then motivate—through a causal-effect-identification example—why it must be addressed head-on rather than assumed away.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , two environments e and e' can differ through several distinct distribution shifts, catalogued in Table 1. Our first observation is that these shifts typically trace back to a single cause: a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that influences $\mathbf{X}$, Y , or both (see Figure 1). Because U drives the observed variables, any shift in $\mathbb{P}(U)$ propagates to distributions over $\mathbf{X}$ and Y alone. Concretely, since $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$ we observe $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$; the same reasoning applies to $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

Table 1: The distribution shifts a single hidden-confounder shift can induce.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

Crucially, most existing OOD methods buy their guarantees by assuming a specific causal structure among U , $\mathbf{X}$, Y that manufactures a convenient invariance. When $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), the factorization $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$ means a joint shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ arises purely from label shift ($\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$), while the conditional $\mathbb{P}(\mathbf{X} | Y)$ stays invariant because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | U)$ [89, 83, 35, 5, 47, 96, 18, 79]. Symmetrically, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), the joint shift is covariate shift ($\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$) while $\mathbb{P}(Y | \mathbf{X})$ stays invariant [70, 16, 80, 11, 28]. The point we want to drive home is that these convenient invariances usually fail, because in practice U causes *both* $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. Under this structure, a single shift in $\mathbb{P}(U)$ triggers label, covariate, conditional-covariate, and concept shifts *simultaneously*. The only invariance that survives is $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ —so-called *stable confounding shift* [85, 4]—and because U is unobserved, it is invisible in the observed data. This is the crux of contribution C1: the invariance that would rescue existing methods is exactly the one hidden confounding hides.

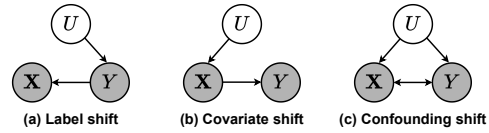


Figure 1: Causal graphs behind the shifts: (a) and (b) yield the convenient invariances existing methods assume, while (c) $U \rightarrow \mathbf{X} \rightarrow Y$ is the confounded case where the joint shift is

Information-theoretic measures. To make these shifts measurable, we follow Federici et al. [22] and quantify them with mutual information: $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$, where E is the environment variable, track shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ respectively. In this language, existing methods assume away one term: label-shift methods set $I(\mathbf{X}; E | Y) = 0$ and covariate-shift methods set $I(Y; E | \mathbf{X}) = 0$. The takeaway of C1, restated

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

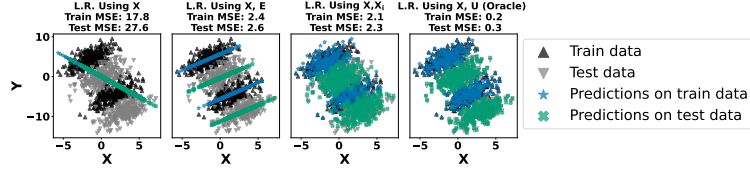


Figure 2: **One picture, three contributions.** Four linear regression (L.R.) models under hidden confounding shift with minimal environment overlap (distant μ_e). (i) Trained on X alone, the model learns the *wrong* X – Y relationship (Simpson’s paradox)—the failure C1 attributes to confounding. (ii) Environment-specific summary statistics of X , denoted E , recover the correct relationship (C2) but are limited in representational power. (iii) An informative covariate X_i proxying for U improves OOD generalization further (C3). (iv) The oracle trained on X and U sets the performance ceiling.

information-theoretically, is that under hidden confounding shift *both* terms are generally nonzero, so neither assumption holds.

A motivating experiment. The experiment in Figure 2 makes the challenge concrete and previews all three technical contributions. The outcome Y depends on X and a hidden confounder U , and environment-specific shifts in the mean of U (denoted μ_e) induce distinct environments. A linear regression trained on X alone infers the *wrong* X – Y relationship (first subplot): Y rises with X within each environment, yet the model captures the opposite trend—a textbook instance of Simpson’s paradox [73]. Dedicated OOD methods (IRM [6], VREX [38], Group DRO [69]) fail the same way. This is the failure that C2 explains and repairs: in § 4 we prove that the optimal strategy is to learn *environment-specific* relationships, which recovers the correct X – Y mapping. The second subplot confirms this—environment-specific summary statistics of X (mean, standard deviation, quantiles) uncover the true relationship, mirroring *backdoor adjustment* in causal effect estimation [58], where environment-specific information acts as a proxy for the confounder. But those statistics have limited representational capacity, so the fix is only partial—setting up C3: as we show in § 4.3, additional informative covariates that proxy for the unobserved confounder push generalization further still (third subplot). An oracle trained on X and U marks the best-possible mean squared error for reference.

4 Impact of hidden confounding shifts on OOD generalization

This section establishes contribution C2: a theoretical account of how hidden confounding shifts reshape OOD generalization. Our vehicle is a decomposition of predictive information (§ 4.1) that we then specialize to the confounded case (§ 4.2), where it reveals that environment-specific relationships are indispensable. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X, Y}[\log \frac{\mathbb{P}(X, Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The goal in OOD generalization is to learn a predictor $(f \circ \phi)$ that performs well on both ID and OOD data. To this end, various objectives have been considered in the literature. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments: $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ where $\mathcal{E}_{tr}$ denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ where β is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ with the constraint that f is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) based methods simply pool data from all training environments and minimize the empirical risk on the pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$ where $\mathcal{D}^e$ denotes observed data in an environment e .

Our analysis centers on a single, task-agnostic objective: the predictive information $I(Y; \hat{Y})$ with $\hat{Y} = (f \circ \phi)(\mathbf{X})$, which measures how informative the prediction is about the true label and thus serves as a natural target across many learning tasks. Studying how hidden confounding shift acts on this quantity is what lets us compare existing objectives on common ground. As building blocks, we define two properties of $\phi(\mathbf{X})$ —*informativeness* and *invariance*—both known to matter for OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $INF(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $CINF(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness measures how much $\phi(\mathbf{X})$ tells us about Y within each environment. Minimizing CINF discards useful signal, and maximizing it can be harmful in settings such as algorithmic fairness, where $\phi(\mathbf{X})$ might exploit sensitive within-environment information. Such bias can be suppressed by minimizing $I(\phi(\mathbf{X}); E)$, but—as we show in § 4.2—this can also cost predictive performance. Setting fairness aside (i.e., when E carries no sensitive information), we adopt the working position that maximizing CINF aids generalization, a claim we substantiate in § 4.1 and that becomes the linchpin of C2.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $VAR(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $INV(\phi(\mathbf{X}), Y, E) = -VAR(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

Invariance quantifies how consistent $\phi(\mathbf{X})$ is across environments for a given label Y . Minimizing invariance risks overfitting by retaining environment-specific information in $\phi(\mathbf{X})$, whereas maximizing it strips out those dependencies and promotes invariant learning—the behavior most existing methods pursue. Extending the measures from § 3, we write $I(\phi(\mathbf{X}); E)$ for *feature shift* and $I(Y; E | \phi(\mathbf{X}))$ for *concept shift*; where context permits, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$. These named terms are exactly the ones our decomposition will expose and rebalance.

4.1 A general decomposition of predictive information

Before stating our first result, we recall the causal-graph machinery it relies on. A causal graph $\mathcal{G}$ consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \leftarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are marginally independent but become conditionally dependent when conditioning on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$. If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$. A path is open if it is not blocked.

With these preliminaries in hand, we turn to the predictive information $I(Y; \hat{Y})$, where the predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are based on a learned representation $\phi(\mathbf{X})$. We model the causal relationships among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as in Figure 3. Here, U is a hidden confounding variable. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$, $\hat{Y}$ is redundant for reasoning about Y . That is, $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. E denotes an environment variable that captures shifts in $\mathbb{P}(U)$. That is, between any two environments, there is a shift in $\mathbb{P}(U)$. We first present a more general decomposition of predictive information without explicit consideration of how U influences $\mathbf{X}, Y$. A formal treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

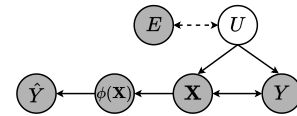


Figure 3: The causal model behind our decomposition. The bi-directed arrow between $\mathbf{X}$ and Y signals that some covariates of $\mathbf{X}$ cause Y while others are caused by Y .

Proposition 4.1. *For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature*

243 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

244 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 245 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

246 Proofs are in Appendix § A. The value of Proposition 4.1 is that it reduces the entire zoo of OOD
 247 methods to a single question—which terms of Equation (1) does a method push up or down?—and
 248 thereby sets the stage for C2. For instance, IRM [6] enforces that a fixed classifier f remains
 249 the same across different environments. This is directly related to maximizing $-I(\phi(\mathbf{X}); E|Y)$
 250 because if $\phi(\mathbf{X})$ were to carry extra environment-specific information conditional on Y , the fixed
 251 classifier would no longer be optimal in all environments. DANN [23] and the independence
 252 criterion for fair classification [22] aims to minimize environment-specific information in $\phi(\mathbf{X})$
 253 by minimizing $I(\phi(\mathbf{X}); E)$. In the context of domain adaptation, certain properties of features
 254 are learned invariant to domains [82, 90]. While minimizing $I(\phi(\mathbf{X}); E)$ may be good for fair
 255 classification, it may degrade predictive performance [98, 33, 22]. This can be seen from Equation (1),
 256 where minimizing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to obtain
 257 $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. By enforcing this equality, E becomes independent of the pair
 258 $(\phi(\mathbf{X}), Y)$. Consequently, $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus,
 259 although CDAN does not explicitly include $I(Y; E | \phi(\mathbf{X}))$ in its loss, its joint-distribution alignment
 260 objective effectively drives this conditional mutual information to zero. While $I(Y; E)$ is a constant,
 261 there exist methods that are robust to shifts in the distribution of labels [69, 95].

262 So far Equation (1) is agnostic to how U influences $\mathbf{X}$ and Y ; making that structure explicit tells
 263 us which terms to attack. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, so minimizing
 264 $I(\phi(\mathbf{X}); E | Y)$ is principled. If instead $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, which
 265 motivates minimizing $I(Y; E | \phi(\mathbf{X}))$. The hard—and realistic—case is $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$: here
 266 the terms interact non-trivially and no existing recipe is obviously optimal. Resolving this case is
 267 precisely contribution C2, which we deliver in the next section.

268 4.2 Predictive information under hidden confounding shift

269 We now resolve the hard case and, with it, prove C2. To see how a hidden confounder U that causes
 270 both $\mathbf{X}$ and Y acts on the predictive information, we treat *two special cases* of the $\mathbf{X}$ – Y relationship:
 271 (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, we obtain the inequalities in (2) because conditioning on
 272 Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y which results
 273 in additional information flow from $\phi(\mathbf{X})$ to E via Y . In contrast, conditioning on $\phi(\mathbf{X})$ partially
 274 blocks the information flow from Y to E at the node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information
 275 about $\mathbf{X}$.

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \quad \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

276 Similarly, when $Y \rightarrow \mathbf{X}$, we obtain the inequalities in (3) because conditioning on Y blocks
 277 the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path
 278 $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ because $\phi(\mathbf{X})$ is the child of $\mathbf{X}$ and conditioning on
 279 the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \quad \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

280 Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

281 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 282 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 283 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

284 Proposition 4.2 handles the two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, but the first dominates practice:
 285 11 of the 16 datasets in recent benchmarks [57, 50, 24] follow $\mathbf{X} \rightarrow Y$, and even in the remaining
 286 5, far fewer covariates are caused by Y than cause it. The consequence—stated as the takeaway
 287 below—is the heart of C2.

Takeaway (C2). Under hidden confounding shift, whenever either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$ (i.e., (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ or (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$), the decomposition (4) collapses: the *variation, label shift, feature shift, and concept shift* terms of (1) all vanish. Predictive information is then maximized by maximizing a single difference, *conditional informativeness* – *residual*—an inherently *environment-specific* quantity. Enforcing invariance alone cannot achieve this, which is why invariance is necessary but not sufficient.

288

289 **Implication 1—maximizing conditional informativeness.** C2 immediately rationalizes an existing
 290 method and exposes its limits. Prashant et al. [62] recently proposed a mixture-of-experts (MoE)
 291 model for OOD generalization under hidden confounding shift, with one expert per hidden-confounder
 292 assignment; each expert maximizes performance within the environment defined by a particular
 293 confounder value. This is precisely maximizing conditional informativeness $I(\phi(\mathbf{X}); Y | E)$, so
 294 Equation (4) supplies the theoretical justification for MoE-type models. Just as importantly, it flags
 295 what such models leave open: the method of Prashant et al. [62] assumes overlapping confounder
 296 support, $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, and discrete-valued proxies—assumptions our next box
 297 shows are not always needed.

Figure 2 shows that, even without confounding overlap and with a *continuous*-valued proxy, a linear regression recovers the correct covariate–target relationship across environments once given environment-specific information. This concretely motivates the open problem C2 surfaces: OOD generalization methods that stay effective when confounding support is limited and the confounder is continuous-valued.

298

299 **Implication 2—minimizing the residual.** The second lever in C2 is the residual $I(\phi(\mathbf{X}); Y | \hat{Y})$, which quantifies how much information $\phi(\mathbf{X})$ carries about Y beyond what the prediction $\hat{Y}$
 300 already captures. The residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ is related to the expected cross-entropy loss as:
 301 $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$ is known as the
 302 *refinement error* that measures the model’s ability to distinguish between classes. Minimizing the
 303 residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus involves two complementary strategies. First, one can directly
 304 minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$,
 305 i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can minimize the refinement error $H(Y | \hat{Y})$, for instance,
 306 through training techniques that minimize refinement error for accurate predictions before minimizing
 307 the calibration error [10].

310 4.3 Impact of additional informative variables

311 We now turn to contribution C3. Section 4.2 gave the optimal strategy when the confounder is entirely
 312 unobserved; in practice we often have *proxies* for it [4, 85, 62] that can stand in for the hidden
 313 confounder. This subsection quantifies, through the same decomposition, exactly how adding such
 314 non-causal but informative covariates moves the predictive information.

315 **Definition 4.3** (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,*
 316 *neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are*
 317 *not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

318 Because $U \rightarrow Y$, any covariate informative about U is also informative about Y , and vice versa.
 319 The decomposition (1) says that maximizing $I(Y; \hat{Y})$ requires reducing concept shift $I(Y; E | \phi(\mathbf{X}))$ —which is exactly what Liu et al. [50] observe empirically: collecting additional covariates
 320 $\mathbf{X}_I$ that make $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ more stable across environments improves OOD test accuracy. C3 puts
 321 this observation on a formal footing: we prove that informative variables raise predictive information
 322 by moving specific terms of (1) in their beneficial directions.
 323

Table 2: **Evidence for C1:** every shift type predicted by hidden confounding is present. Mean $\pm$ standard deviation over 10 random subsets of 40,000 samples. One-sample t-tests against a mean of zero ($\alpha = 0.05$) confirm all four measures are significantly nonzero (p-value ≈ 0) in every dataset.

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

Proposition 4.3. If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned}
 (i) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) &> I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\
 (iii) \ I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) &< I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y)
 \end{aligned} \tag{5}$$

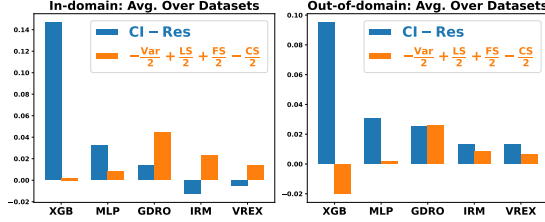
Proposition 4.3 is the formal content of C3: adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii) while reducing concept shift (5.iii)—all favorable for predictive information. The one caveat, which we flag rather than hide, is that it also amplifies variation (5.iv), so dedicated strategies may be needed to keep variation in check.

5 Experimental results

This section delivers contribution C4: empirical confirmation of the three claims above. Our experiments on real-world and synthetic data are organized to test them one by one—(i) that hidden confounding shift is genuinely present in real-world data (evidence for C1), (ii) that the components of the decomposition (1) govern OOD generalization under hidden confounding shift, with conditional informativeness driving accuracy (evidence for C2), and (iii) that informative covariates improve generalization (evidence for C3). We use eight real-world tabular benchmark datasets: *Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*. These datasets and corresponding domain splits are adopted from *TableShift* benchmark [24]. We use ID test and OOD test accuracies to measure the performance of models. We perform experiments on two ERM-based methods: XGBoost (XGB) [15] and multi-layer perceptron (MLP), two domain generalization methods: IRM [6], VREX [38], and one robust learning based method: Group DRO (GDRO) [69]. We present average results over 5 different random hyperparameter choices as described in Appendix B. The effect of different random seeds is statistically insignificant for these datasets [24]. Additional details of the experimental setup are provided in Appendix § B. Results with hyperparameter tuning are presented in Appendix C.

Result 1 (supports C1): hidden confounding shift is real and pervasive. C1 predicts that a hidden confounding shift should leave a distinctive fingerprint—simultaneous shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} \mid Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y \mid \mathbf{X})$. Table 2 shows exactly this signature: label, covariate, conditional-covariate, and concept shift are all significantly nonzero across every real-world dataset, consistent with an underlying confounding shift. Qualitatively, we prompt GPT-4o [1] to name plausible hidden confounders for several benchmarks [24, 57] (Appendix § E); for *food stamps*, for example, unmeasured state-specific *economic policies* plausibly drive both *household income* and *food stamp reciprocity*.

Result 2 (supports C2): conditional informativeness tracks accuracy. Proposition 4.2 predicts that, under hidden confounding shift, predictive information hinges on the single difference *conditional informativeness* – *residual*. Figure 4 (first six datasets with more than one training environment [57, 24]; per-dataset results in Appendix C) confirms this prediction: that difference is positively correlated with both ID and OOD accuracy, and XGB—the best-performing model—attains the largest value of it. Two further observations reinforce C2. First, the theory-predicted collapse actually happens: the orange bars, i.e. the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 -$



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: **Evidence for C2.** The difference *conditional informativeness*—*residual* (plots) is positively correlated with average ID and OOD test accuracy over the eight datasets (table, right). XGB, which maximizes this difference, is the top performer—exactly as Proposition 4.2 predicts.

Table 3: **Evidence for C3.** Sign consistency is high for concept shift (**CS**) across all methods, but the term Proposition 4.2 deems decisive—conditional informativeness (**CI**)—is where XGB excels, matching its top accuracy. **Res**: residual, **Var**: variation, **FS**: feature shift, **C**: causal, **AC**: arguably causal, **A**: all.

	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
Method	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

361 *concept shift* / 2, sit near zero as Proposition 4.2 requires. Second, the relative ID/OOD ranking of
362 methods matches previously reported results [57, 24], so the effect is not an artifact of our setup.

363 **Result 3 (supports C3): informative covariates improve generalization.** We test C3 by measuring
364 what happens as the covariate set grows. Following [57], we use three nested subsets *causal* $\subseteq$
365 *arguably causal* $\subseteq$ *all covariates* (Appendix § B) and track how the terms of (1) move as we enlarge
366 the set. To summarize this cleanly, we introduce the *sign consistency metric*: for each positively
367 weighted term (e.g. conditional informativeness) we count how often it *raises* when accuracy improves,
368 and for each negatively weighted term (e.g. concept shift) how often it *falls*; the metric is the fraction
369 of moves in which a term shifts in its *beneficial* direction, i.e. how reliably it aids generalization
370 (formal definition in Appendix § B). Table 3 bears out C3: both conditional informativeness and
371 concept shift move in their beneficial directions with high consistency. Reducing concept shift is
372 common to all methods, but XGB additionally raises conditional informativeness more than the
373 others—precisely the term C2 identifies as decisive—yielding its extra performance. Per-dataset
374 results are in Appendix § C.

375 A caveat of the real-world test is that we cannot verify whether *all available covariates* truly contain
376 *every* relevant informative variable. We therefore corroborate C3 on synthetic data with a fully known
377 causal structure— $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ are informative covariates
378 caused by the hidden confounder U . Here the prediction is unambiguous, and it holds: adding $\mathbf{X}_I$
379 raises conditional informativeness and feature shift while lowering concept shift (Figure 5), with
380 the variation term staying at zero as expected in this simple setting. Full data-generation details and
381 further results appear in Appendix § D.

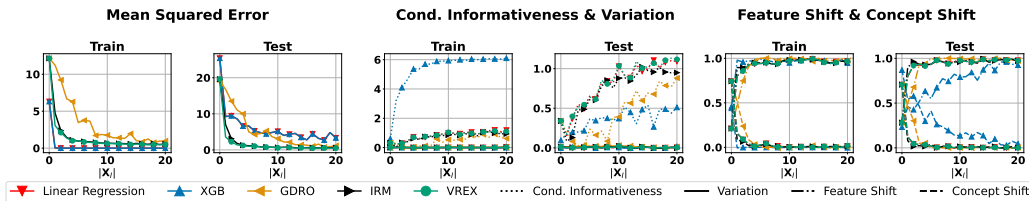


Figure 5: **Evidence for C3 on synthetic data.** Adding more proxy variables $\mathbf{X}_I$ of U that are informative about Y reduces MSE, raises conditional informativeness and feature shift, and lowers concept shift—the exact term movements Proposition 4.3 predicts.

6 Discussion, limitations, and future work

We set out to explain why ERM and all-covariate models succeed under real-world distribution shifts, and our four contributions together supply that explanation. **(C1)** Hidden confounding shift is the mechanism: it entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), so treating any one in isolation is doomed—and no finite set of observed covariates or proxies can fully dissolve the resulting ambiguity. **(C2)** Our decomposition of predictive information (Equation (1)) isolates the factors limiting OOD generalization and shows that, under confounding shift, learning environment-specific input–output mappings beats enforcing a single invariant predictor (Equation (4)); invariance is necessary but not sufficient. **(C3)** Proxies for hidden confounders provably raise predictive information, which is why adding informative, non-causal covariates helps. **(C4)** Experiments on eight real-world benchmarks and synthetic data confirm all three. By making the inevitability of hidden confounding explicit—and by pinpointing conditional informativeness as the quantity to maximize—our work lays a foundation for more robust OOD methods. We see four open challenges: (i) algorithms effective under limited confounder overlap, (ii) quantifying the cost–accuracy trade-off of acquiring informative covariates, (iii) handling entangled shifts without untestable proxy assumptions, and (iv) modeling the residual ambiguity from unobserved confounders to inspire new OOD-robust paradigms.

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677 Appendix

678 A Proofs of propositions

679 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 680 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 681 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

682 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 683 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

684 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 685 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 686 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 687 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

688 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

689 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

690 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 691 expansion instead of $\phi(\mathbf{X})$. $\square$

692 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 693 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 694 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

695 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 696 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

697 **Case 1:** $\mathbf{X} \rightarrow Y$. In this case, we have the following inequality because conditioning on Y opens the
 698 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

699 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 700 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 701 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

702 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

703 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

704 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

705 **Case 2:** $Y \rightarrow \mathbf{X}$. In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 706 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 707 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

708 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 709 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

710 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

711 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

712 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

713 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 714 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

715 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 716 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

717 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
718 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
719 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
720 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

721 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
722 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
723 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
724 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
725 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

726 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

727 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

728 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
729 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
730 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
731 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
732 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

733 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
734 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
735 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
736 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
737 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

738 B Experimental setup

739 This section documents the experimental setup underlying contribution C4. The impact of random
740 seeds is statistically insignificant in many settings [24]. For the results in the main paper, we report
741 the mean and standard deviation across five random hyperparameter values (sampled from their
742 respective domains in Table B1) for each dataset-method combination. Since we evaluate models
743 on ID test and OOD test data, to evaluate mutual information terms, we use the nonparametric
744 entropy estimation toolbox [37, 77, 78]. Following [24], we evaluate mutual information using both
745 in-distribution (ID) test data (from training domains) and out-of-distribution (OOD) test data (from
746 test domains).

747 For consistency we draw 20,000 random sam-
748 ples from each ID and OOD test set, though
749 some datasets are smaller. Sample size has lit-
750 tle effect on the estimated mutual-information
751 terms—an important robustness check for C4:
752 as Figure B1 shows, the terms of Equation (1)
753 for the XGB representation on *income* stay sta-
754 ble as sample size varies. Features $\phi(\mathbf{X})$ come
755 from the layer preceding the classification head;
756 for XGBoost we combine output margins and
757 SHAP values. All experiments ran on a single
758 NVIDIA RTX A6000 GPU.

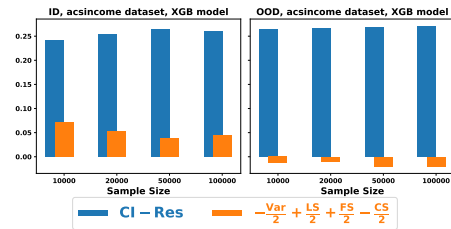


Figure B1: Evaluated terms of predictive information are relatively consistent across various sample sizes on the *income* dataset.

759 For the informative-covariate experiments behind C3 and C4, we follow the categorization of Nastl
760 and Hardt [57], partitioning covariates into *causal*, *arguably causal*, and *all covariates*. Causal
761 covariates have clearly established influence on the target, with well-supported directionality and
762 unlikely reverse causation. However, strict reliance on causal variables risks omitting relevant
763 parents due to knowledge gaps. Arguably causal covariates have uncertain causal relationships,
764 meeting at least one of: (1) being known causal, (2) having plausible but potentially bidirectional
765 influence, or (3) likely (but unconfirmed) causal effect. These groups approximate true causal
766 parents based on available knowledge, though relationships may be confounded. Some datasets

Table B1: Hyperparameters and their possible values considered in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

767 additionally contain *anti-causal* covariates where the target likely affects them but not vice versa.
768 The *all covariates* set includes all observed variables regardless of causal status. The sign consistency
769 metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
770 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
771 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

772 Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates
773 and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and

test data. $\mu_s^{tr/te}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{causal, arguably\ causal, all\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section reports per-dataset results that reinforce C2. Figures C2, C3 break down, for each dataset, the terms *conditional informativeness – residual* and *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2* (aggregated in the main paper’s Figure 4, left). Two points stand out. Consistent with Proposition 4.1, the sum of the blue and orange terms correlates positively with overall performance. More tellingly for C2, hidden confounding shift makes the difference *conditional informativeness – residual* the *dominant* contributor to predictive information, dwarfing *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*—which is exactly why maximizing conditional informativeness is the right objective under confounding shift.

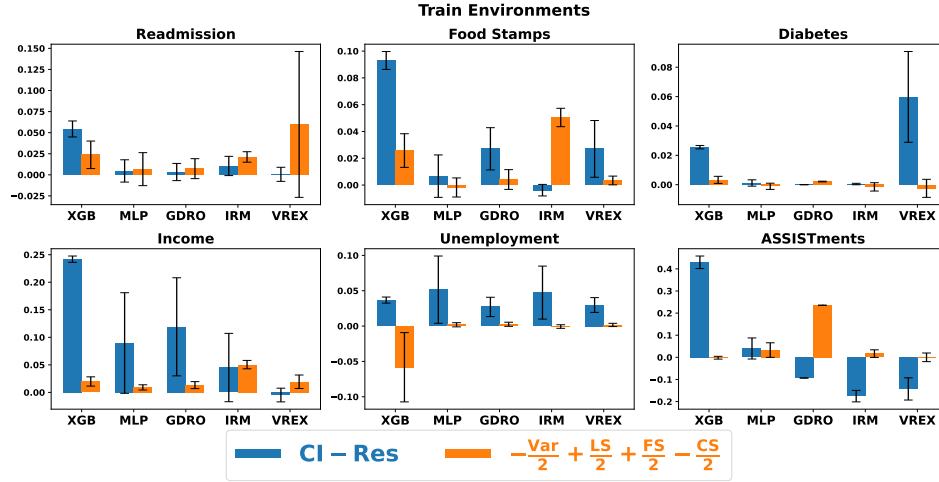


Figure C2: Dataset specific results on ID data.

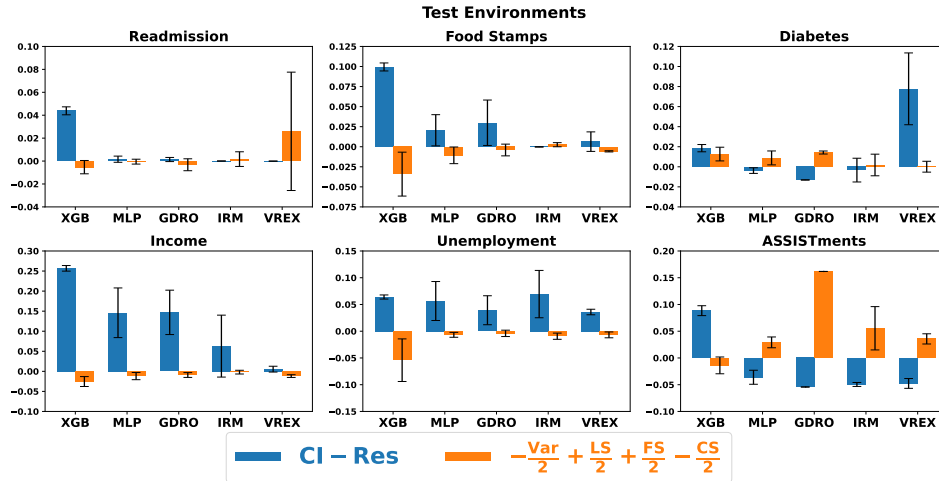


Figure C3: Dataset-specific results on OOD data.

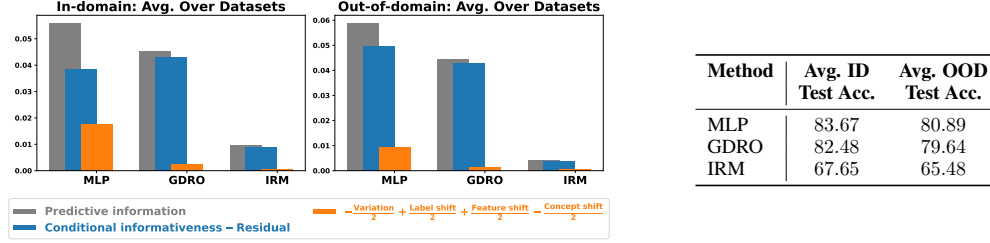


Figure C4: **C2 holds under hyperparameter tuning.** The difference *conditional informativeness* – *residual* (plots) is positively correlated with average ID and OOD test accuracy over the five datasets (table, right), and it contributes more to overall predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

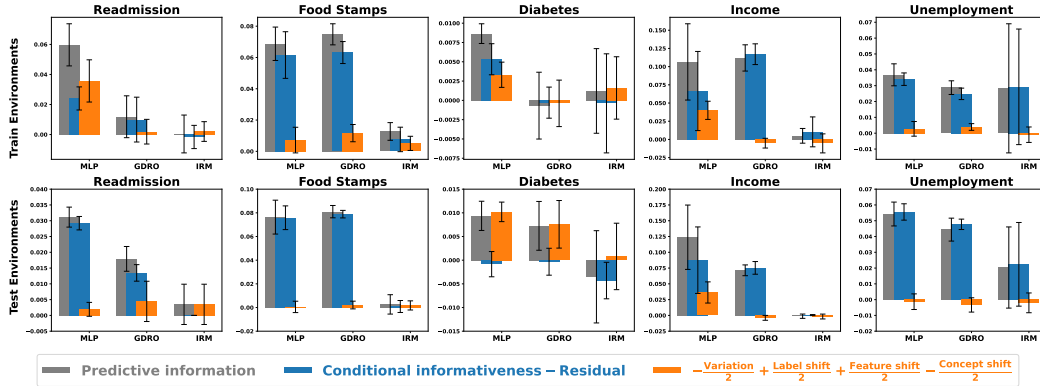


Figure C5: Dataset specific results. Comparison of the terms in predictive information.

Table C2: Comparison of ID and OOD test accuracies of models.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9

	Income		Unemployment	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00

Figures C6- C15 provide a fine-grained, per-method and per-dataset view of the decomposition terms in (1), and they make the mechanism behind C2 visible. The second and third columns (*variation* and *feature shift*) are comparable in magnitude, so by Equation 1 their difference nearly cancels; the same holds for the fourth and fifth columns (*label shift* and *concept shift*). What remains—in almost every setting—is the difference *conditional informativeness* – *residual* as the dominant contributor to predictive information, exactly as the collapse in Proposition 4.2 predicts.

Robustness to hyperparameter tuning (C2 again). To rule out that Result 2 is an artifact of our default hyperparameters, we repeat the analysis with tuning. We grid-search hyperparameters and select the best by *ID* validation accuracy (not OOD), running 1500 experiments (5 datasets $\times$ 3 models $\times$ 100 hyperparameter choices), each averaged over 5 random seeds even where seed effects are negligible. The conclusion is unchanged (Figure C4): consistent with Proposition 4.1, the sum of the *blue* and *orange* terms tracks performance, and under hidden confounding shift the

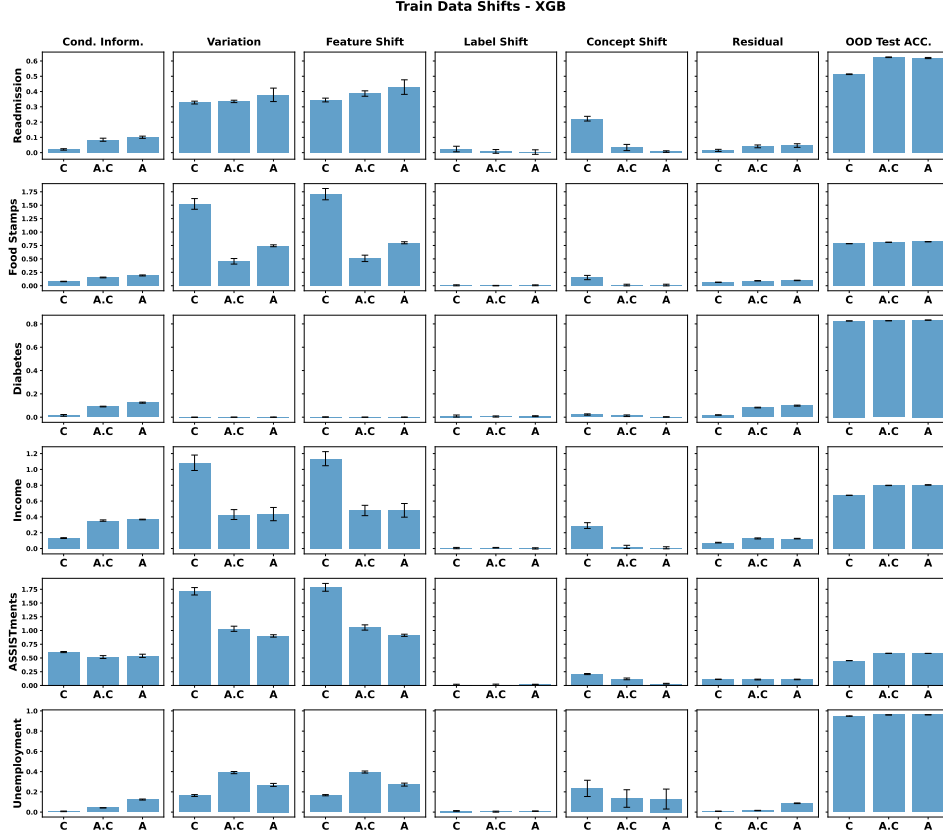


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

799 difference *conditional informativeness* – *residual* again dominates – *variation* / 2 + *label shift* / 2 +
800 *feature shift* / 2 – *concept shift* / 2. Per-dataset results (Figure C5, Table C2) show the same trend.

801 **Limitations of the predictive-information analysis.** In the interest of full disclosure, we note where
802 our C2/C3 analysis stops short. Predictive information $I(Y; \hat{Y})$ measures statistical dependence, not
803 correctness: a binary classifier that systematically flips the label (e.g. $\hat{Y} = 1$ when $Y = 0$) attains the
804 maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation, yet scores 0% accuracy, because mutual
805 information captures *reduction in uncertainty* rather than accuracy. The individual terms carry similar
806 caveats—conditional informativeness $I(\phi(X); Y|E)$ can mask harmful variation $I(\phi(X); E|Y)$, and
807 feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ via spurious correlations. These caveats notwithstanding,
808 the decomposition remains a reliable diagnostic for reasonably performant models—as our
809 experiments show—because it distinguishes predictive information rooted in generalizable patterns
810 from that rooted in environment-specific artifacts, enabling targeted, interpretable improvements.

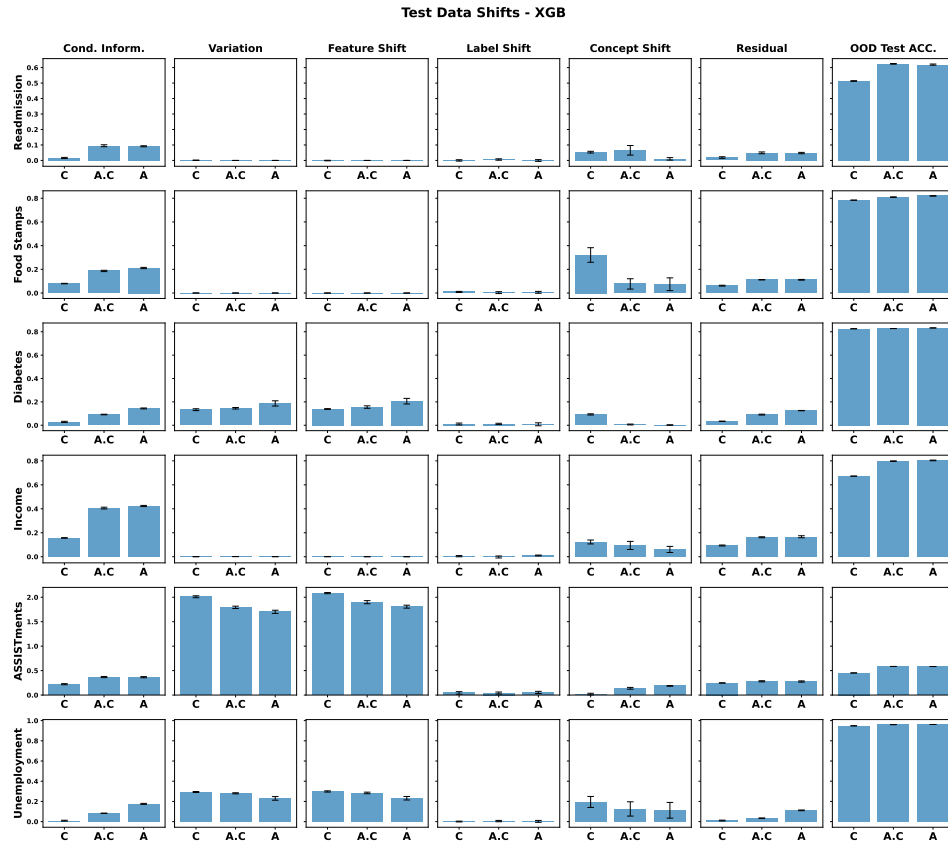


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

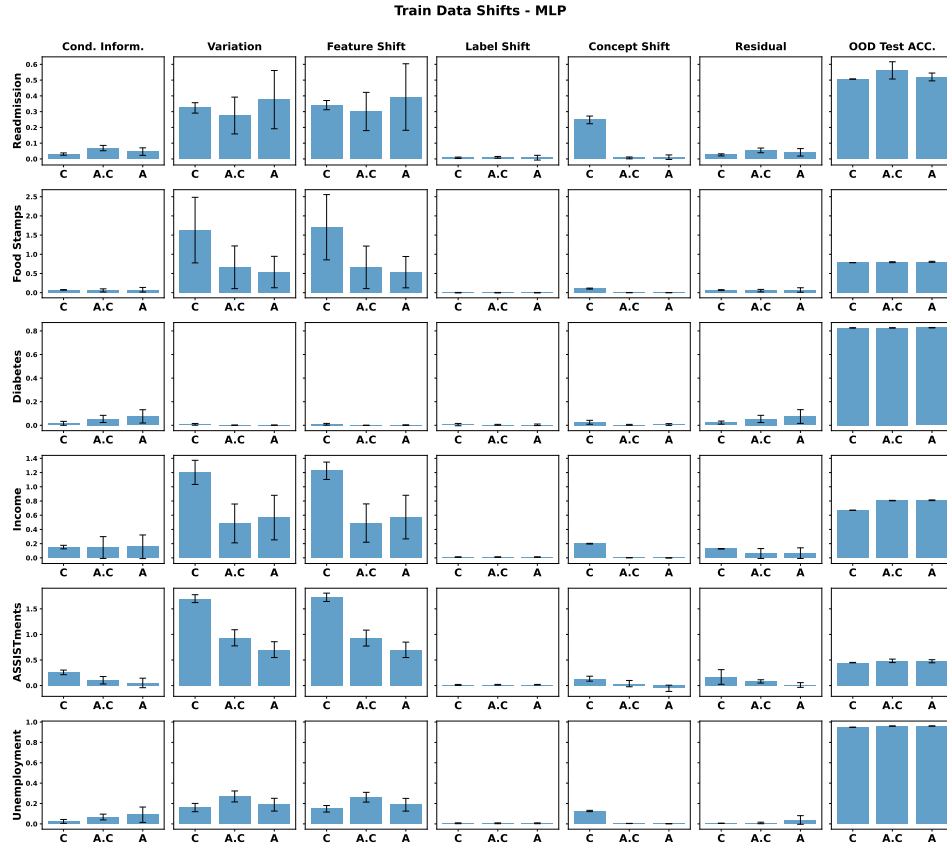


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

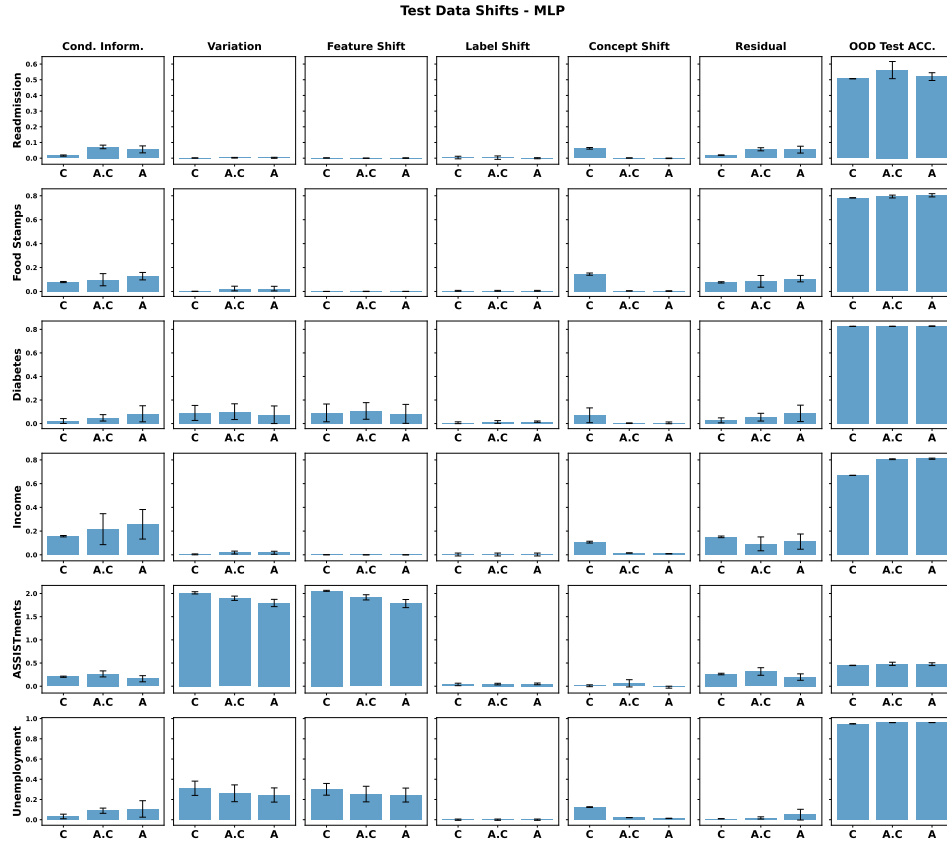


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

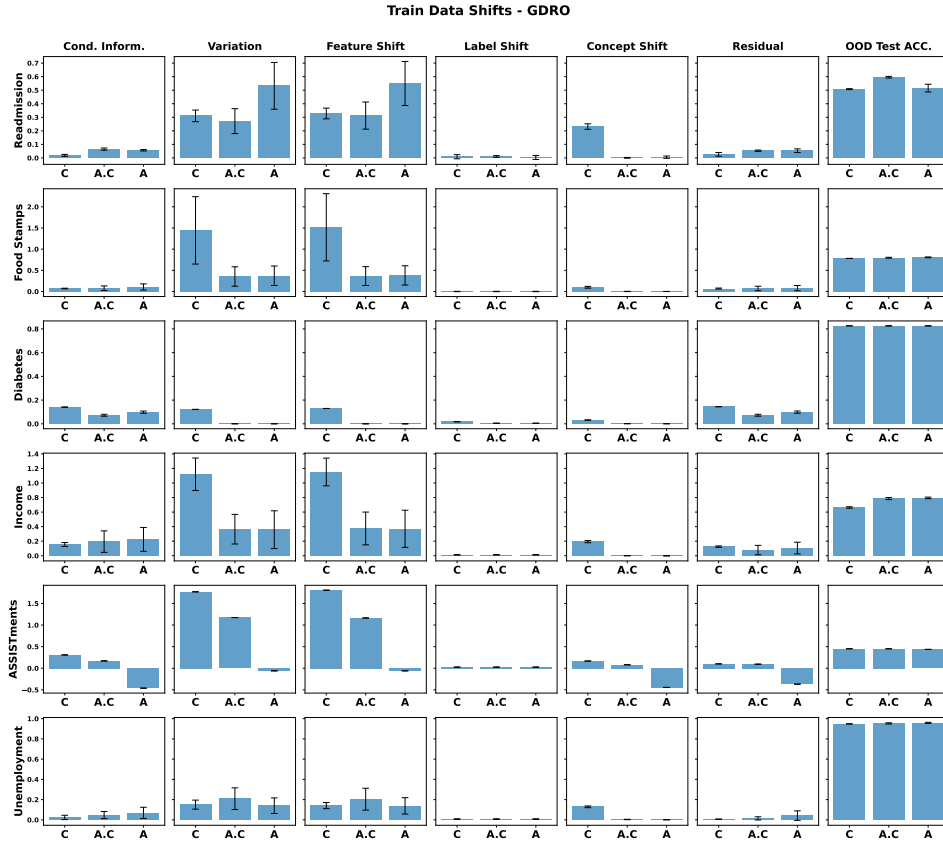


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

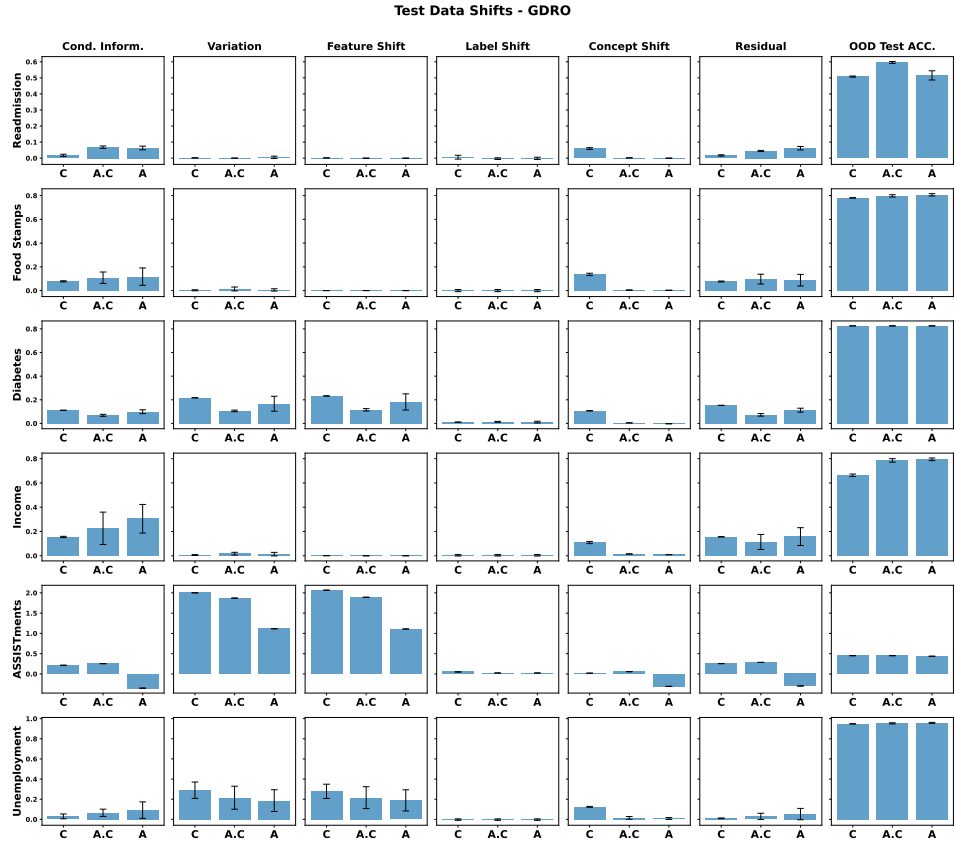


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

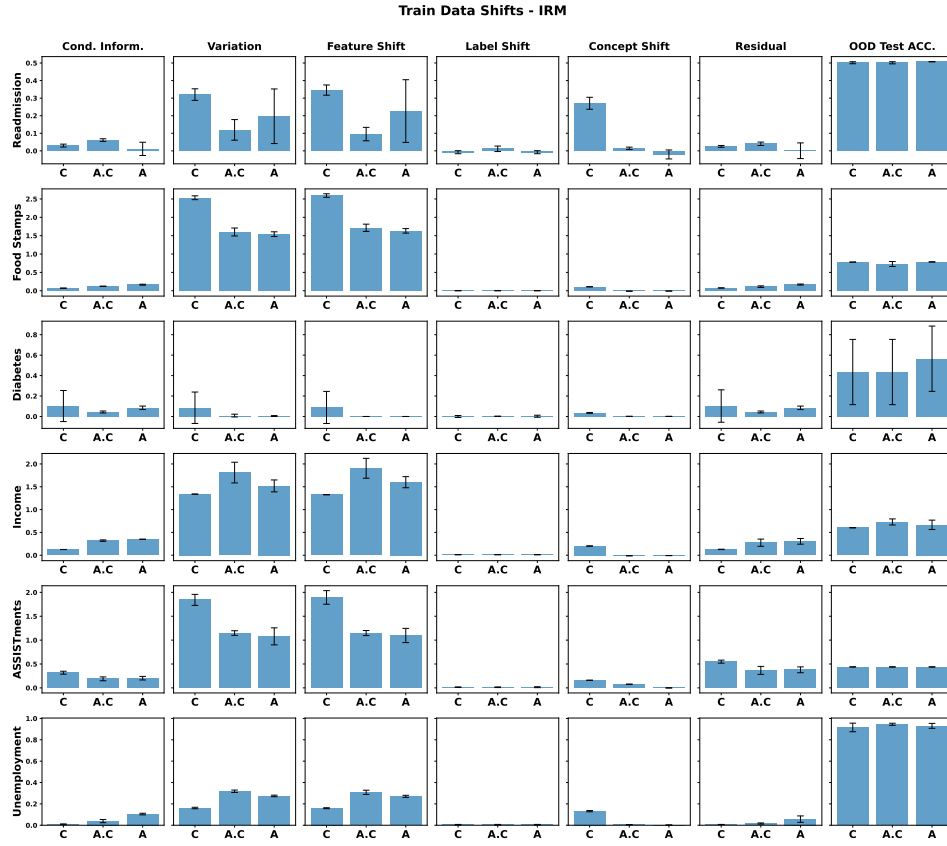


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

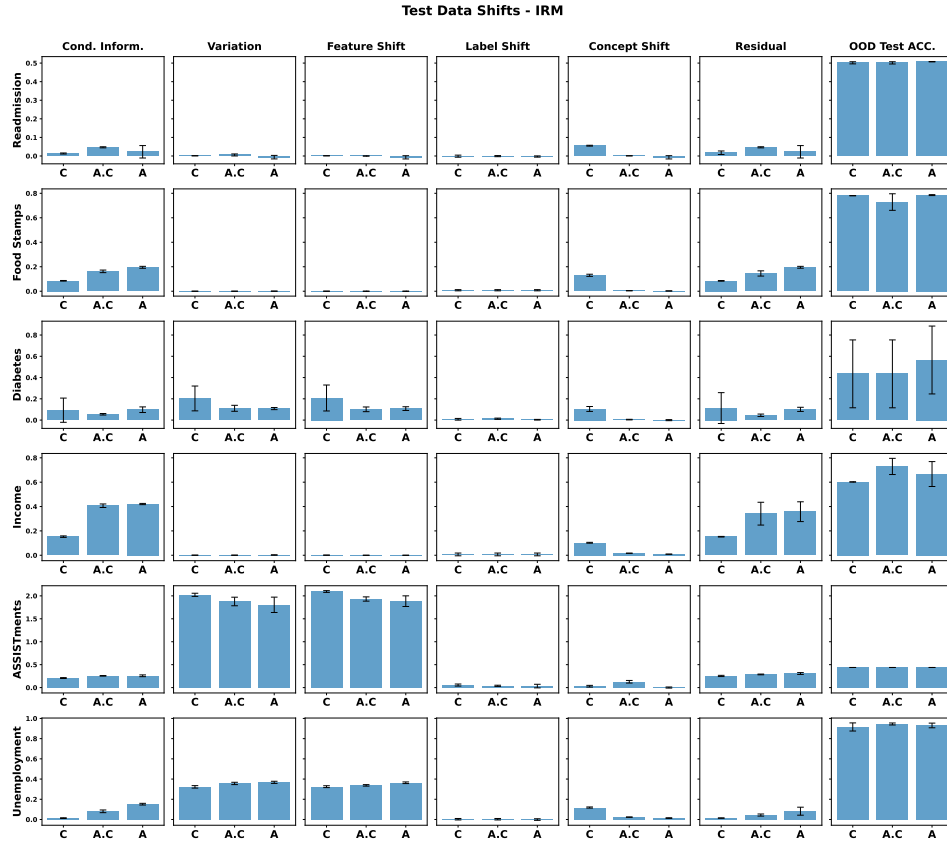


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

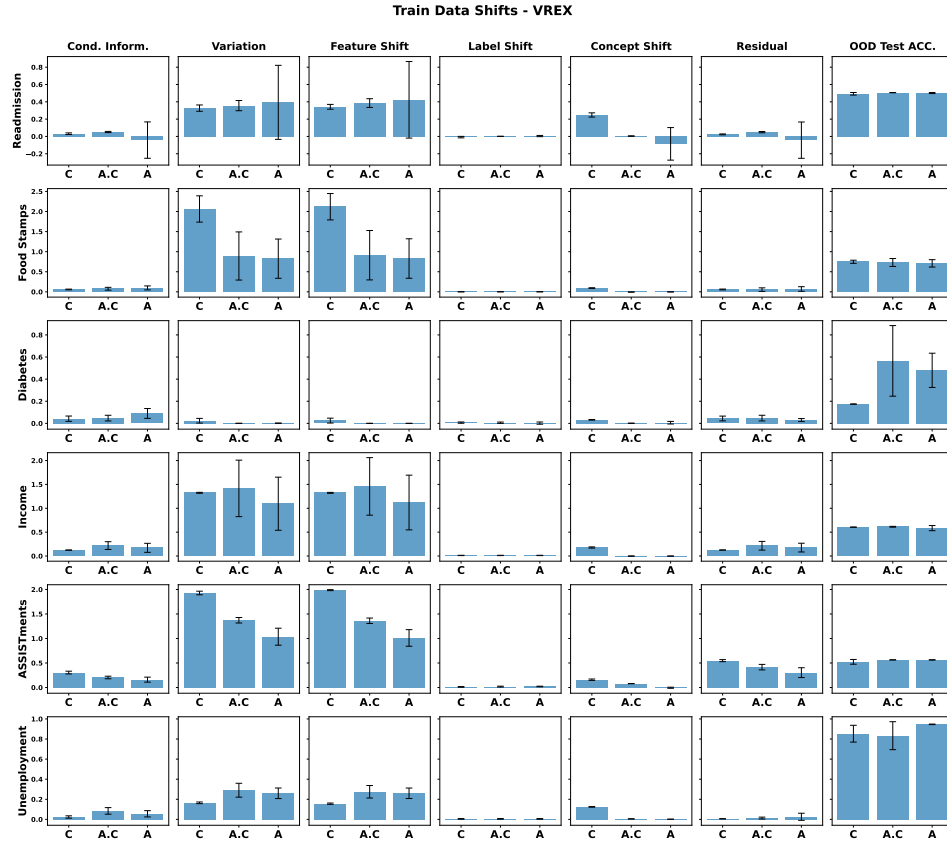


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

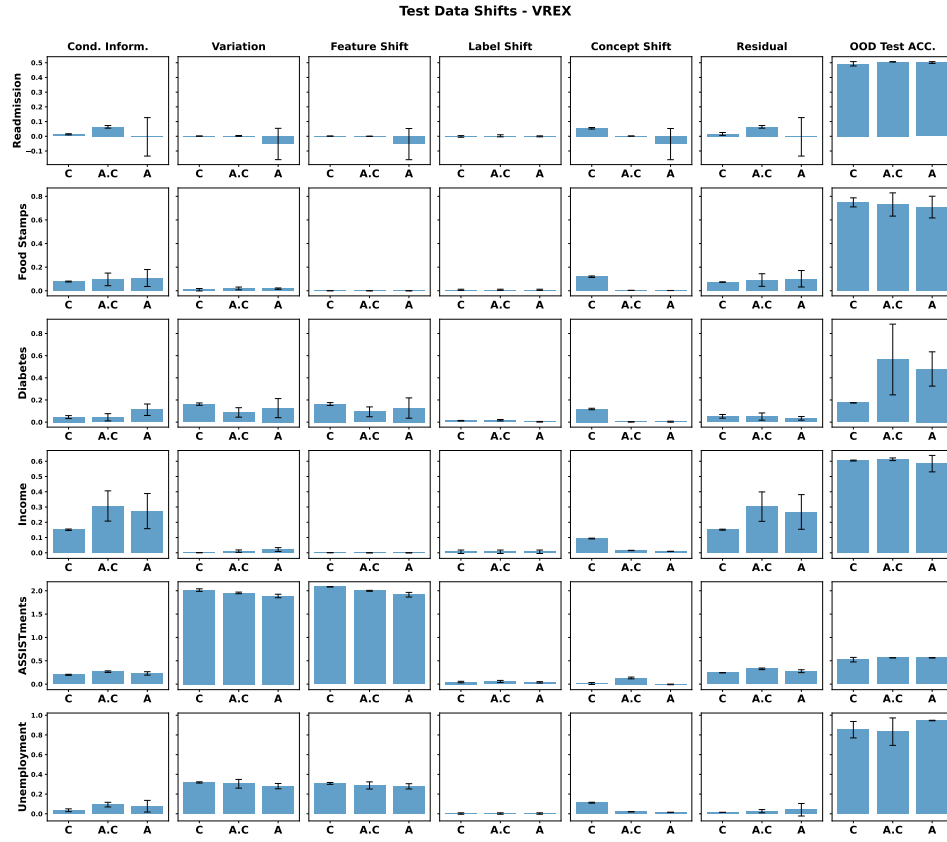


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

D Results on synthetic datasets

These synthetic experiments extend the empirical support for C3 (and probe the open problems raised by C2) by exploiting full control over the data-generating process. We investigate three questions: (i) how different informativeness criteria affect the gains from proxy covariates, (ii) how the degree of confounder overlap across environments affects performance, and (iii) how the balance of causal versus anti-causal variables shapes it. The data-generating process follows the structural equations in (12), with the causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is an unobserved confounder, X is an observed covariate, Y is the target variable, and $\mathbf{X}_I$ represents additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

We explore different functional forms for f_X , f_i , and f_Y , with the corresponding results presented in our analysis. Domain shifts are induced by systematically varying the environment-specific mean parameter μ_u^e of the confounder distribution. This allows us to examine how different degrees of distributional shift affect model performance across environments.

Informativeness vs. accuracy (C3). We begin with linear structural equations: $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$. For these experiments $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test environments. For the results presented in Figure 5 of the main paper, we consider one hidden confounding variable with $|\mathbf{X}_I| = 20$ informative covariates. As we increase the number of informative covariates from 0 to 20, we observe: (i) a reduction in mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while (iv) decreased concept shift. We extend these findings by examining two additional informativeness criteria. First, we analyze the case with one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$) across varying noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$. Figure D16 demonstrates that as noise decreases from 2.0 to 0.0, we observe: (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, alongside (iv) decreased concept shift. Consistent with other synthetic experiments, the variation term remains zero throughout. Finally, we investigate a setting with 20 hidden confounders and corresponding 20 informative covariates, using modified structural equations $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17 shows that as we introduce informative covariates corresponding to hidden confounders, we again observe: (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while (iv) decreased concept shift. These experiments demonstrate how different informativeness criteria help in achieving OOD generalization.

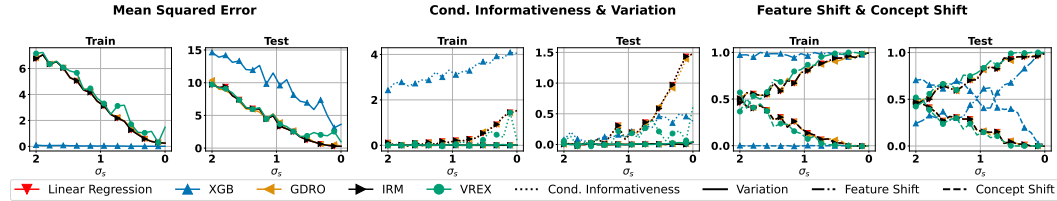


Figure D16: Adding proxy variable with low noise helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

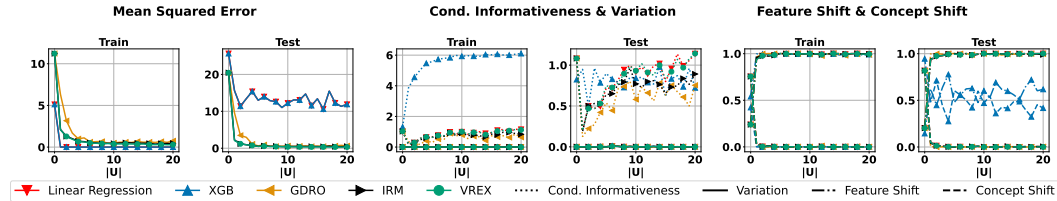


Figure D17: Adding more proxy variables $\mathbf{X}_I$ of a set of hidden confounding variables $\mathbf{U}$ that are informative to Y helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

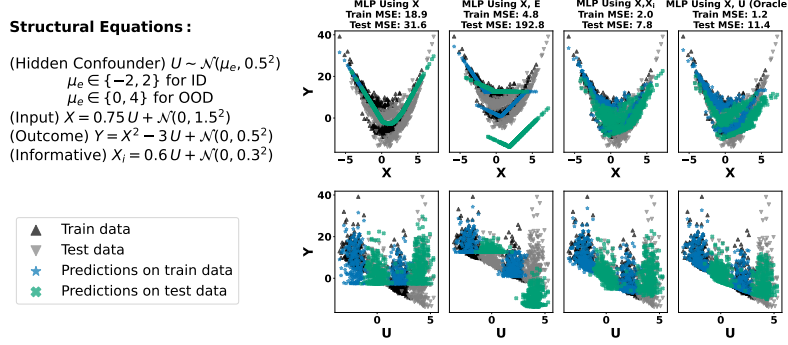


Figure D18: Performance of MLP on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

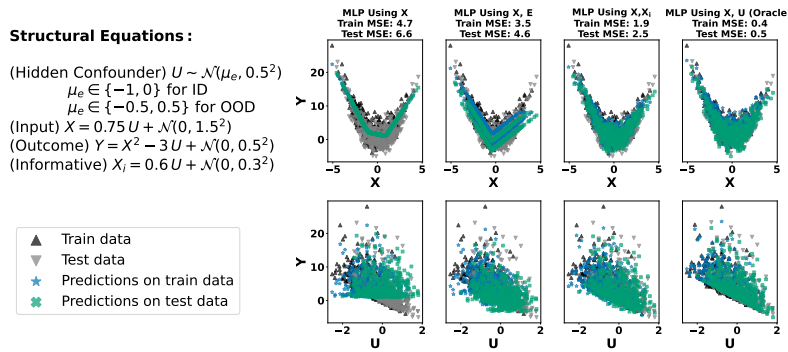


Figure D19: Performance of MLP on the high overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

842 **Overlapping confounder support vs. accuracy.** This experiment attacks the open problem C2
843 surfaced: whether the common overlapping-support assumption of Prashant et al. [62] is truly
844 necessary. Having shown that the correct relationships are recoverable in synthetic *linear* data when
845 observed covariates carry enough information about the confounder, we now push to *nonlinear*
846 data with MLP and XGB under both low and high confounding-support conditions (Figures D18–
847 D21). Under low overlap (distant μ_e between ID and OOD data), models trained on X alone learn
848 an inadequate global function that cannot separate environments; environment-specific summary
849 statistics help (C2), but the largest gains again come from additional informative covariates (C3). For
850 completeness we include oracle results and U – Y scatter plots that expose the learned input–output
851 mappings. As expected, high overlap is easier; the more consequential finding is that the low-overlap
852 regime—largely unstudied and a direct challenge to the common-support assumption—remains
853 tractable.

854 **Number of causal vs. anti-causal variables.** This experiment probes the very phenomenon that
855 motivated the paper. Recent benchmarks [24] report that baselines (XGB, MLP) match or beat OOD
856 methods (GDRO, IRM, VREX) on real-world data, coinciding with the prevalent $\mathbf{X} \rightarrow Y$ structure
857 in tabular data. To ask whether the causal structure itself explains baseline superiority, we sweep
858 the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$
859 (e.g. $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). Baselines stay resilient even under $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$
860 (Figure D22), achieving superior PMA-OOD scores [24] (fraction of maximum OOD accuracy),
861 though no single method dominates universally (Figures D23, D24). We evaluate: (1) ERM-
862 based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer,
863 NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM,
864 IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain
865 Mixup [90, 91]); and (3) Robust optimization methods (DRO [42], GroupDRO [69], Label DRO,
866 Adversarial Label DRO [94]).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

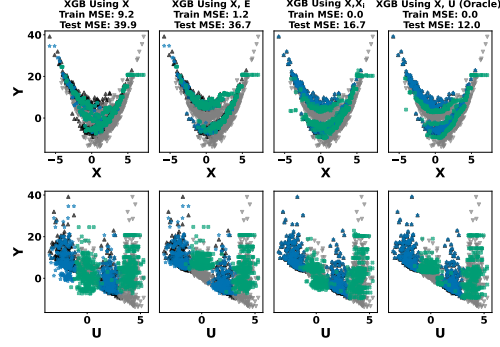


Figure D20: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

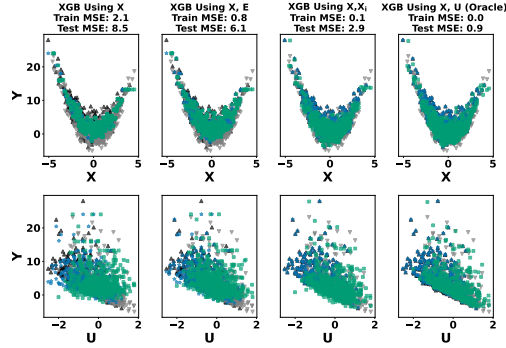


Figure D21: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_o|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

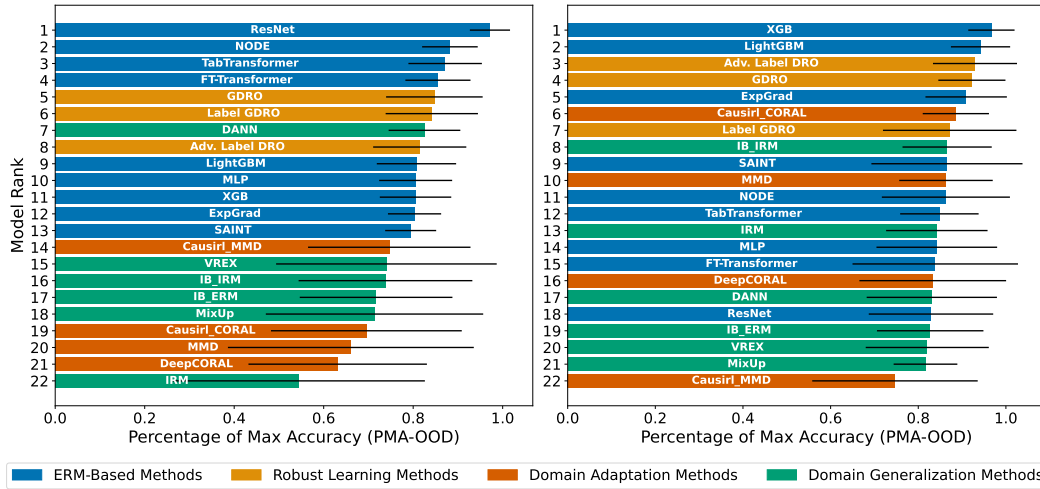


Figure D22: ERM-based methods achieve a higher percentage of maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, reproducing—now in a controlled setting—the real-world pattern of [24] that motivates this paper.

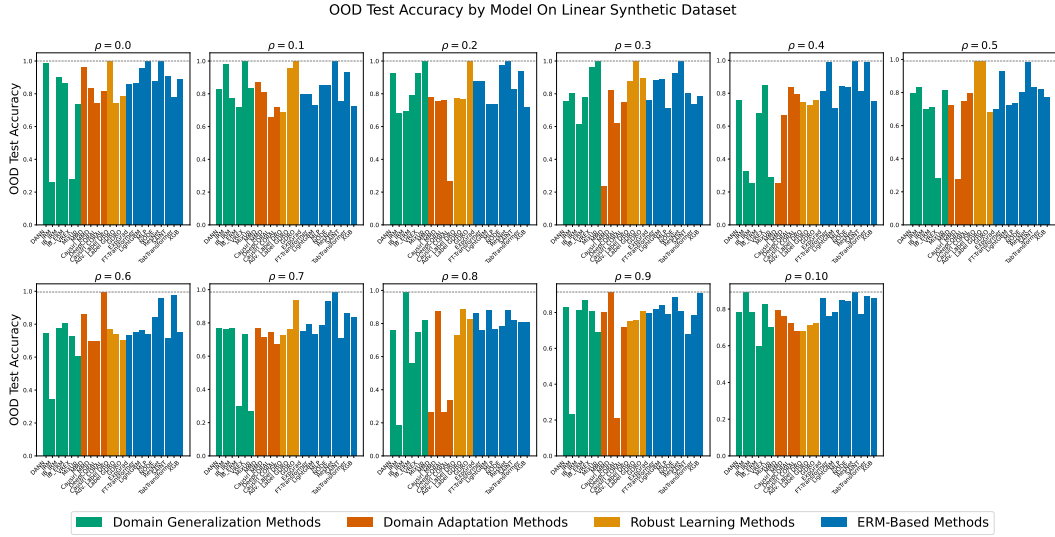


Figure D23: OOD test accuracy on the linear setting per model for different values of ρ .

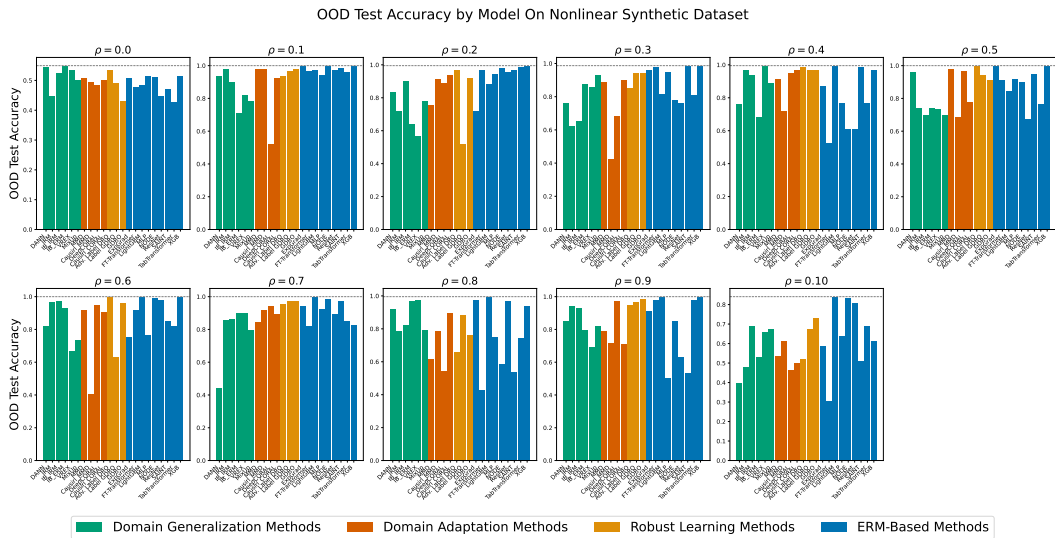


Figure D24: OOD test accuracy on the non-linear setting per model for different values of ρ

867 E Qualitative analysis of hidden confounding shift in real-world datasets

868 This section provides qualitative support for C1: that hidden confounding shifts genuinely arise in
 869 real-world data. Tables E3-E18 list plausible hidden confounders influencing observed covariates
 870 across several benchmarks [57] (dataset details in [24, 57]). We elicit these with GPT4o [1] and stress
 871 that the lists are illustrative, not exhaustive. Together with the quantitative signature in Table 2, they
 872 reinforce the case that hidden confounding shifts exist and matter across domains.

873 **Prompt:** We use the prompt below to query GPT4o to get possible hidden confounders.

874 For a target variable, you will be given lists of causal, arguably
 875 causal, anti-causal, and spurious covariates. Also, you will be provided
 876 with environment variables such that different datasets are collected in
 877 environments induced by these environment variables. Provide possible
 878 hidden confounders that cause any of the target, causal, arguably causal,
 879 spurious, and anti-causal covariates. That is, the hidden confounders
 880 should influence the distribution of their children in different domains.
 881 Ensure hidden confounders are not a part of the given variables, and your
 882 explanation should include how hidden confounders influence their children
 883 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of in-stitution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.